

Spotify Track Popularity Prediction

Introduction

Problem Definition

Music streaming platforms such as Spotify rely heavily on understanding user engagement to improve recommendations, playlist curation, and promotional strategies. One of the strongest indicators of engagement is **track popularity**, measured on a scale from **0 to 100**.

Objective

The objective of this project is to **predict the popularity of a Spotify track** using available metadata related to:

- Track characteristics
- Artist attributes
- Album information
- Temporal features

This is formulated as a **supervised regression problem**.

Why This Problem Matters

- Real-world **business relevance** (recommendation systems, music marketing)
- Excellent example of **tabular regression**, ideal for ML model comparison
- Enables comparison between **traditional ML models and Deep Learning (ANN)**
- Highlights the importance of **feature engineering and explainability**

Dataset Overview

Data Source

The dataset consists of Spotify track metadata collected from the Spotify platform.

Target Variable

- **track_popularity** (continuous, range: 0–100)

Feature Types

Feature Type	Examples
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Numerical	Track duration, artist popularity
Categorical	Artist genre, album type
Temporal	Release year

Each row represents a **single track**, making the dataset suitable for supervised learning.

Data Preprocessing

The dataset underwent extensive preprocessing to ensure reliability and modeling readiness.

Preprocessing Steps Completed

- ✓ Missing values handled
- ✓ Duplicate records removed
- ✓ Date fields parsed correctly
- ✓ Invalid numeric ranges filtered
- ✓ Categorical values cleaned and standardized

Outcome

A **clean, high-quality dataset** with no data leakage or structural issues, ready for exploratory analysis and modeling.

Exploratory Data Analysis (EDA)

Distribution of Target Variable

Key Observations

- The popularity distribution is **right-skewed**
- Most tracks fall into **low to medium popularity**
- Very popular tracks are **rare**

Implication

This imbalance increases prediction difficulty and penalizes models that fail to capture non-linear patterns.

Correlation Analysis

Findings

- **Artist popularity** has the strongest positive correlation with track popularity
- Track duration shows weak correlation

- No severe multicollinearity among numerical features

Implication

Linear models are safe to apply, but may struggle with non-linear dependencies.

Genre Frequency Analysis

Observations

- Dataset is **genre-imbalanced**
- Mainstream genres dominate the data

Implication

Model predictions may be biased toward popular genres, limiting generalization to niche music styles.

Feature Engineering (Critical Section – 20%)

Feature Selection (Bias & Leakage Control)

Non-predictive and leakage-prone features were removed:

- Track name
- Artist name
- Album identifiers
- Release date fields

Rationale

- Prevents memorization
- Improves generalization
- Ensures model fairness

Feature Creation

Track Age

A new feature was engineered:

track_age = 2025 – release_year

Why this matters ?

- Captures temporal influence on popularity
- Older tracks may accumulate popularity over time
- More informative than raw release year

Categorical Encoding

Categorical variables were encoded using **Label Encoding**.

Justification

- Works well with tree-based models
- Compatible with neural networks
- Avoids high-dimensional sparsity caused by one-hot encoding

Feature Scaling

All numerical features were standardized using **StandardScaler**.

Why scaling is essential

- Required for distance-based models (KNN, SVR)
- Improves ANN convergence
- Ensures fair feature contribution

Machine Learning Models

Evaluation Metrics

Metric	Purpose
RMSE	Penalizes large prediction errors
R^2	Measures explained variance

Models Trained & Results

1. Linear Regression

- High RMSE
- Underfits the data
- Cannot capture non-linear relationships

2. K-Nearest Neighbors (KNN)

- Sensitive to scaling

- Performs poorly in high-dimensional space
- Suffers from curse of dimensionality

3. *Decision Tree*

- Captures non-linearity
- Overfits without depth control

4. *Random Forest*

- Best-performing ML model
- Robust to noise
- Strong bias–variance balance

5. *Support Vector Regression (SVR)*

- Powerful non-linear model
- Computationally expensive
- Highly sensitive to hyperparameters

Structured Model Comparison

Model	Performance	Interpretation
Linear Regression	Poor	Underfitting
KNN	Poor	Dimensionality issues
Decision Tree	Medium	Overfitting
Random Forest	Best	Strong generalization
SVR	Medium	Requires careful tuning

Hyperparameter Tuning

Grid Search on Random Forest

Tuned parameters:

- Number of trees
- Maximum tree depth

Results

- Lower RMSE after tuning

- Reduced overfitting
- Improved test-set generalization

Key Insight

Hyperparameter tuning is essential for extracting full model potential.

Model Explainability

Feature Importance (Random Forest)

Top Influential Features

1. Artist popularity
2. Track age
3. Genre

SHAP Global Explanation

Insights

- Artist popularity dominates predictions
- Older tracks tend to receive higher predicted popularity
- Genre has moderate but consistent influence

SHAP Local Explanation

Purpose

- Explains individual predictions
- Shows why a specific track received a particular popularity score

This increases **model transparency and trustworthiness**.

Deep Learning – Artificial Neural Network (ANN)

Architecture

- Input layer
- Two hidden layers with ReLU activation

- Dropout layers for regularization
- Output layer with linear activation

Training Observations

- Smooth convergence
- Validation loss closely follows training loss
- Dropout successfully reduces overfitting

Performance Comparison

- ANN performs competitively
- Does **not outperform Random Forest**

Expected Outcome

Tree-based models often outperform neural networks on structured tabular data.

Limitations & Critical Discussion

- Popularity influenced by **external factors** (marketing, playlists, virality)
- Genre encoding is simplistic
- Dataset biased toward mainstream music
- ANN complexity does not guarantee better performance on tabular data

Conclusion

Final Takeaways

- **Random Forest** is the best-performing and most reliable model
- ANN is competitive but not superior
- Feature engineering has a larger impact than model complexity
- Explainability (SHAP) is crucial for trust in predictions

Final Statement

This project demonstrates that **well-engineered features + interpretable models** outperform overly complex architectures in real-world tabular regression problems.